Czech Republic: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- The Czech Republic is most exposed in upstream gas, downstream gas and upstream electric grid, where its Energy Security is higher than only 13%, 16% and 28% of the other economies scored. In upstream gas, import needs and the absence of recorded reserves pull the score down.[1]
- Its strongest openings are in midstream coal, midstream nuclear and midstream oil, where its Economic Opportunity is higher than 98%, 97% and 96% of the other economies scored.[1]
- On the paper's weighting, the Strategic Index ranks downstream solar 1st among the Czech Republic's own sectors; it is among the sectors with the most policy.[1]
- The Partnership Strength Index rates Germany highest for the Czech Republic, above all as an export market, and it is also the Czech Republic's 1st trade partner. China, its 3rd trade partner, ranks 43rd on Partnership Strength.[1,2]

1. Energy security risks and why

7 of the Czech Republic's 30 placed sectors, those where both scores can be compared with other economies, sit in the two low-energy-security corners. Energy Security measures how well a country can rely on what it needs. Compared with other economies, its Energy Security is lowest in upstream gas (0.26), downstream gas (0.40) and upstream electric grid (0.33), all in the Upgrade corner, which pairs high opportunity with low energy security. These scores are higher than only 13%, 16% and 28% of the other 97, 98 and 93 economies scored in each sector.[1]

In upstream gas, the score is low mainly because of import needs and reserves. Gas production minus consumption was -0.25 EJ in 2025 (production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately). Its reserves score is 0.00 against a median of 0.18: no proved gas reserves are recorded, which the index counts as zero. For context, its largest suppliers in this sector in 2024 were Norway (65.2% of imports worth USD 2.88 bn) and Russia (31.2%).[1,2,3,4]

In downstream gas, energy access pulls the score down: gas consumption was 24.2 GJ per person in 2025, after a change of -16.1%. In upstream electric grid, trade pulls it down: on the trade deficit in these goods as a share of the economy, the Czech Republic scores higher than only 17% of the other economies.[1,3]

The high-priority sectors, taken from the Upgrade corner in Strategic Index order, are downstream solar, upstream electric grid and upstream electric vehicles. In downstream solar, which scores 0.47 on Energy Security, the reserves category, here solar resource potential, pulls the score down: on that measure it scores higher than only 12% of the other economies.[1]

2. Economic opportunities and why

27 of the 30 placed sectors sit in the two high-opportunity corners, where Economic Opportunity, how well placed a country is to create value, is at or above the sector median. Compared with other economies, its Economic Opportunity is highest in midstream coal, midstream nuclear and midstream oil, all in the Grow corner. These scores are higher than 98%, 97% and 96% of the other 154, 153 and 155 economies scored.[1]

In all of them, trade and cost competitiveness push the score up. Exports of these goods were USD 519 m, USD 266 m and USD 1.43 bn in 2024. On input costs of labour and capital, the Czech Republic scores higher than 88%, 90% and 91% of the other economies. For context, the largest markets for its midstream coal exports in 2024 were Germany (28.9%), Poland (12.4%) and Turkiye (7.5%).[1,2,5]

Among the high-priority sectors, downstream solar has Economic Opportunity higher than 82% of the other 147 economies scored. Cost competitiveness and investment push it up: on input costs it scores higher than 95% of the other economies, and on annual investment higher than 76%.[1]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by opportunity, security need, policy activity, stage and low-emission value. On the paper's weighting, it puts downstream solar 1st, upstream wind 2nd, upstream solar 3rd, upstream batteries 4th and upstream electric grid 5th among the Czech Republic's own sectors. The score of downstream solar is 0.66, 0.016 ahead of the next sector; the next are almost tied (gaps of 0.004, 0.000, 0.003 and 0.000). Policy activity carries its lead: its policy part adds 0.200, against 0.176 for opportunity and 0.158 for security need. On data-based priority, the Strategic Index without that part, it would fall to the middle third. It sits in the Upgrade corner, which the paper's framework associates with building domestic capability.[1]

The Czech Republic's allied pick, its top upstream or midstream sector after a screen for a recorded resource base, is midstream nuclear (0.63), almost tied with midstream solar (0.007 apart). The screen removes upstream wind, upstream solar, upstream batteries and upstream electric grid.[6]

Policy activity counts the International Energy Agency's Policies and Measures records and the Climate Action Tracker rating, which favours low-emission technologies over coal, oil and gas. Country Insights splits priority and policy activity into thirds; both are high in midstream nuclear, midstream batteries and midstream solar. No sector pairs high priority with the least policy; among high priorities, policy is weakest in upstream green hydrogen (4th on priority, 19th on policy). The reverse, low priority with the most policy, applies to downstream green hydrogen. The Czech Republic carries the Climate Action Tracker rating Insufficient (November 2024) and has 127 Policies and Measures records.[1,7,8]

4. Partnership priorities and why

The Partnership Strength Index rates partners as suppliers (Friendshore), as export markets (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. From the Czech Republic's viewpoint it ranks Germany 1st, the United Kingdom 2nd, the United States 3rd, the Slovak Republic 4th and Poland 5th of 193 partners. Germany and the United Kingdom score best as an export market; the best sectors for Germany include midstream electric grid, upstream solar, midstream wind, midstream coal and midstream batteries. The United States, the Slovak Republic and Poland score best as a reliable supplier.[1]

Germany is also the Czech Republic's 1st trade partner on average across its energy-goods baskets, Poland its 2nd, the Slovak Republic its 4th, the United Kingdom its 5th and the United States its 6th. Each of them is a priority partner with strong ties today, shares membership of NATO with the Czech Republic and sits in the same bloc in UN votes. By contrast, China, its 3rd trade partner, ranks 43rd on Partnership Strength, and Russia, its 7th, ranks 82nd. In UN votes, the Czech Republic and each of them sit in opposite blocs.[1,2,9,10]

As a partner to the coalition, the Czech Republic ranks 10th of 192 economies on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads production of each upstream and midstream sector across coalition members, its largest derived share is 1.3% of midstream coal.[1,6]

5. Data notes and caveats

Paper baseline: 12 October 2026. Scores compare the Czech Republic with other economies in the same sector; they are not measures of risk or profit in absolute terms. All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings. Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average. Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order. Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up. In the allied network design the European Union is one producer; the Czech Republic's shares are derived from its weight within the EU. The Climate Action Tracker rates the EU as a whole; the Czech Republic carries the EU's rating. The Czech Republic's counts leave out the EU's own 438 International Energy Agency policy records. Trade figures count trade with other EU members.

Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
2. United Nations, UN Comtrade Database, HS6 bilateral trade (all reporters), 2020 and 2024; reconciled by the authors (pulled 2026-10-07 to 2026-10-09); attribution; https://comtradeplus.un.org/.
3. Energy Institute, Statistical Review of World Energy 2026 (data to 2025); attribution; https://www.energyinst.org/statistical-review.
4. U.S. Energy Information Administration, International Energy Statistics, bulk file INTL.zip: crude oil incl. lease condensate, NGPL, dry natural gas and coal production; petroleum and other liquids, dry natural gas and coal consumption, bulk file dated 2026-10-03 (oil production to 2025; gas and coal to 2024); open; https://www.eia.gov/international/data/world.
5. United Nations, UN Comtrade Database, exports to World by reporter, 2024 (2024 data); attribution; https://comtradeplus.un.org/.
6. Allied network design (the paper's model design of coalition production shares).
7. International Energy Agency, Policies and Measures Database (export of 2026-09-30); attribution; https://www.iea.org/policies.
8. Climate Action Tracker, country ratings (ratings updated 2022-2025); attribution; https://climateactiontracker.org/.
9. Economic-security agreements: G7, NATO, Indo-Pacific Economic Framework, Minerals Security Partnership (and Forum), U.S. free-trade agreements (compiled by the authors from official membership lists).
10. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.
